Information Security

Assignment 4

Total Marks: 70	**Date:**
Name:	**ID:**

Attempt All Questions only Hand written Solution will be graded

Read Carefully

Soft Copy Submission: Through Portal

Late Submission Policy: 10% Deduction /24 Hours.

Instructions: (5 marks will be deducted for not following the instructions)

- Attempt all questions in sequence. Attach this title page as a front page of assignment.
- Assignment should be handwritten/printed on A4 sized page. **(No pages from register please.)**
- Submit soft copy of solved assignment on the Google Classroom **(should be submitted before deadline)**

Total Marks: 20 (70% Assignment + 30% Evaluation)

Answers Clarity: Justify your answer where needed.

Question # 1: **(10 Marks)**

Given the following 64-bit plaintext and 56-bit key, perform **one round of DES encryption** and show all the steps involved.

- **Plaintext**: `0110100001101001011011000110111101101111011101000110110001101100`
- **Key**: `0001001100110100010101110111100110011011101111001101111111110001`

Tasks:

1. Apply the **Initial Permutation (IP)** on the given plaintext.
2. Split the result into two 32-bit halves, **L0** and **R0**.
3. Perform the **Expansion Permutation (E/P)** on the right half **R0**.
4. XOR the result of the **E/P** with the subkey **K1** (first subkey generated from the given key).

Initial Permutation (IP) Table:

58	**50**	**42**	**34**	**26**	**18**	**10**	**2**	**60**	**52**	**44**	**36**	**28**	**20**	**12**	**4**
62	54	46	38	30	22	14	6	61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7	59	51	43	35	27	19	11	3

Expansion Permutation (E/P) Table (for 32-bit half block):

32	**1**	**2**	**3**	**4**	**5**	**4**	**5**	**6**	**7**	**8**	**9**	**8**	**9**	**10**	**11**
12	13	14	15	16	17	16	17	18	19	20	21	20	21	22	23
24	25	26	27	28	29	28	29	30	31	32	1	32	31	30	29

Question # 2: **(10 Marks)**

Given the key **"0F1571C947D9E859"** (in hexadecimal), perform the following operations:

1. Convert the given hexadecimal key into its **56-bit binary** equivalent.
2. Apply **PC1 (Permutation Choice 1)** to the 56-bit key to split it into two 28-bit halves: **C0** and **D0**.
3. Perform a **left shift** on both halves by 1 bit to generate **C1** and **D1**.
4. Apply **PC2 (Permutation Choice 2)** to generate the first subkey **K1**.
5. Show the resulting subkey **K1** after the **PC2** permutation.

PC1 (Permutation Choice 1) Table:

57	**49**	**41**	**33**	**25**	**17**	**9**	**1**	**58**	**50**	**42**	**34**	**26**	**18**	**10**	**2**
59	51	43	35	27	19	11	3	60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6	61	53	45	37	29	21	13	5

PC2 (Permutation Choice 2) Table:

14	**17**	**11**	**24**	**1**	**5**	**3**	**28**	**15**	**6**	**21**	**10**	**23**	**19**	**12**	**4**
26	8	16	7	27	20	13	2	41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56	34	53	46	42	50	36	29	32

Question # 3: **(50 Marks)**

Perform the following AES operations step by step on the given 128-bit plaintext block. Show all working clearly.

Given plaintext block in hexadecimal:

```
32 43 F6 A8 88 5A 30 8D 31 31 98 A2 E0 37 07 34
```

and the round key:

```
2B 7E 15 16 28 AE D2 A6 AB F7 15 88 09 CF 4F 3C
```

Tasks:

a) Arrange the given plaintext into the AES **4 × 4 state matrix** using column-major order. **[5 marks]**

b) Arrange the given round key into the AES **4 × 4 key matrix** using column-major order. **[5 marks]**

c) Perform the **AddRoundKey** operation by XORing the state matrix with the key matrix. Show the XOR calculation for at least four bytes. **[10 marks]**

d) Apply the **ShiftRows** operation on the resulting matrix. **[10 marks]**

e) For the first column of the matrix after ShiftRows, calculate the first byte of the **MixColumns** result using:

$$(02 \cdot S_0) \oplus (03 \cdot S_1) \oplus (01 \cdot S_2) \oplus (01 \cdot S_3)$$

Show the Galois Field multiplication and XOR steps clearly. **[15 marks]**

f) Briefly explain why AES uses multiple transformations such as AddRoundKey, ShiftRows, and MixColumns instead of only XOR encryption. **[5 marks]**

Good Luck 😊